

Analysis Tutorial 8

Problem 22

Let $N = \{f_n > f_{n+1} \text{ for some } n \in \mathbb{N}\}$
and $f_n \rightarrow f$ on $\Omega \setminus N$ (either to a
finite value or ∞). f_n converges pointwisely
 μ -almost everywhere to f . By MCT,
 f is measurable on $\Omega \setminus N$ (and $\lim_{n \rightarrow \infty} \int_{\Omega \setminus N} f_n d\mu$).
Set $f \equiv 0$ on N so f is measurable
on Ω , thus f is μ -a.e. finite

$$\begin{aligned} \text{Now } \int_{\Omega} f^+ d\mu &= \underbrace{\int_N f^+ d\mu}_{=0} + \underbrace{\int_{\Omega \setminus N} f^+ d\mu}_{= \lim_{n \rightarrow \infty} \int_{\Omega \setminus N} f_n^+ d\mu} \\ &\quad \int_{\Omega \setminus N} f_n^+ d\mu < \infty \end{aligned}$$

Similarly $\int_{\Omega} f^- d\mu < \infty$ so f is integrable,
and $\int_{\Omega} f d\mu = \lim_{n \rightarrow \infty} \int_{\Omega \setminus N} f_n d\mu$
 $= \lim_{n \rightarrow \infty} \int_{\Omega \setminus N} f_n^+ d\mu - \int_N f_n^+ d\mu$
 $= \lim_{n \rightarrow \infty} \int_{\Omega} f_n d\mu$

Problem 23

Define $g_m := \sum_{n=1}^m f_n$

Then $g_m^+ = \sum_{n=1}^m f_n^+$ and is increasing,
with $\int g_m^+ d\mu = \sum_{n=1}^m \int_{\Omega} f_n^+ d\mu \leq \sum_{n=1}^{\infty} \int_{\Omega} |f_n| d\mu < \infty$

So by (22), there is some measurable,
 μ -a.e. finite function g^+ with
 $\int_{\Omega} g^+ d\mu = \lim_{m \rightarrow \infty} \int_{\Omega} g_m^+ d\mu$

Similarly, we get $\int_{\Omega} g^- d\mu = \lim_{m \rightarrow \infty} \int_{\Omega} g_m^- d\mu$
So $\lim_{m \rightarrow \infty} \int_{\Omega} g_m d\mu = \int_{\Omega} (g^+ - g^-) d\mu$
 $= \sum_{n=1}^{\infty} \int_{\Omega} f_n d\mu$ μ -a.e. finite

Problem 24

(a) $\emptyset \in \mathcal{A}_1$ and $\emptyset \in \mathcal{A}_2$ so $\emptyset \times \emptyset = \emptyset \in S$

Now let $A_1 \times A_2, B_1 \times B_2 \in S$

$$\begin{aligned} \text{Then } (A_1 \times A_2) \cap (B_1 \times B_2) \\ = \underbrace{(A_1 \cap B_1)}_{\in S_1} \times \underbrace{(A_2 \cap B_2)}_{\in S_2} \in S \end{aligned}$$

$$\begin{aligned} \text{Also, } A_1 \setminus B_1 &= \sum_{i=1}^n C_i \text{ for } C_i \in S_1, \text{ p.d.} \\ A_2 \setminus B_2 &= \sum_{i=1}^m D_i \text{ for } D_i \in S_2, \text{ p.d.} \end{aligned}$$

$$\begin{aligned} \text{Now } (A_1 \times A_2) \setminus (B_1 \times B_2) \\ &= (A_1 \setminus B_1) \times (A_2 \setminus B_2) \\ &= \left(\sum_{i=1}^n C_i \right) \times \left(\sum_{j=1}^m D_j \right) \\ &= \sum_{i=1}^n \sum_{j=1}^m (C_i \times D_j) \in S^+ \end{aligned}$$

So S is a semi-ring on $\Omega_1 \times \Omega_2$

(b) Let $\Omega_1 = \{1, 2\}$, $\mathcal{R}_1 = \{\emptyset, \{1\}, \{2\}\}$
 $\Omega_2 = \{1, 2\}$, $\mathcal{R}_2 = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$

$$\text{Then } A = \{1\} \times \{1, 2\} = \{(1, 1), (1, 2)\} \in \mathcal{R}$$

$$A_2 = \{2\} \times \{1, 2\} = \{(2, 1), (2, 2)\} \in \mathcal{R}$$

But $A \cup A_2 = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$ which we cannot express as $R_1 \times R_2$

for $R_1 \in \mathcal{R}_1$, $R_2 \in \mathcal{R}_2$ since

$$|R_1| \leq 1, |R_2| \leq 2 \text{ so } |R_1 \times R_2| \leq 2 < 4$$